

Machine Learning Task Scheme : A-Z



Abstractive QA

BT : Question Answering

TT : Question Answering

Abstractive question answering is the task of generating an answer from context that correctly answers the question.

Acceptability Classification

BT : Text Classification

TT : Text Classification

Acceptability classification is the task of predicting sentence acceptability based on grammatical correctness.

Any-to-Any

EX : Demo example: An application to chat with an any-to-any (image & text) model.

Dataset example: A dataset with multiple modality input and output pairs.

Model example: Strong model that can take in video, audio, image, text and output text and natural speech.

Library example: transformers.

Any-to-any models can understand two or more modalities and output two or more modalities.

Audio Classification

EX : Library example: speechbrain.

Demo example: An application that can classify music into different genre.

Dataset example: A benchmark of 10 different audio tasks.

Model example: An easy-to-use model for command recognition.

NT : Audio Language Identification

Audio Intent Classification

Audio Emotion Recognition

Audio classification is the task of assigning a label or class to a given audio. It can be used for recognizing which command a user is giving or the emotion of a statement, as well as identifying a speaker.

Audio Emotion Recognition

BT : Audio Classification

TT : Audio Classification

Audio emotion recognition is the task of identifying human emotion in an audio sample.

Audio Intent Classification

BT : Audio Classification

TT : Audio Classification

Audio intent classification is the task of deducing the specific purpose or objective behind a user's input in a conversational context from audio.

Audio Language Identification

BT : Audio Classification

TT : Audio Classification

Audio language identification is the task of determining language from an audio sample.

Audio Text-to-Text

EX : Dataset example: A dataset containing audio conversations with question-answer pairs.

Library example: transformers.

Demo example: A space that takes input as both audio and text and generates answers.

Model example: A lightweight model that has capabilities of taking both audio and text as inputs and generating responses.

Audio-text-to-text models take both an audio clip and a text prompt as input, and generate natural language text as output. These models can answer questions about spoken content, summarize meetings, analyze music, or interpret speech beyond simple transcription. They are useful for applications that combine speech understanding with reasoning or conversation.

Audio-to-Audio

EX : Demo example: An application for speech separation.

Library example: asteroid.

Model example: A speech enhancement model.

Dataset example: 512-element X-vector embeddings of speakers from CMU ARCTIC dataset.

Audio-to-Audio is a family of tasks in which the input is an audio and the output is one or multiple generated audios. Some example tasks are speech enhancement and source separation.

Automatic Speech Recognition

EX : Demo example: A powerful general-purpose speech recognition application.

Model example: A powerful ASR model by OpenAI.

Dataset example: 31,175 hours of multilingual audio-text dataset in 108 languages.

Library example: espnet.

NT : Keyword Spotting

Speaker Identification

Automatic Speech Recognition (ASR), also known as Speech to Text (STT), is the task of transcribing a given audio to text. It has many applications, such as voice user interfaces.

Closed Book QA

BT : Question Answering

Text Generation

TT : Question Answering

Text Generation

Closed book question answering is the task of generating an answer that correctly answers the question with no context provided.

Closed Domain QA

BT : Question Answering

TT : Question Answering

Closed domain question answering is the task of generating or extracting an answer from context that correctly answers the question while being restricted to a specific domain.

Conversational

BT : Text Generation

TT : Text Generation

Conversational models refer to technologies, such as chatbots or virtual agents, that users can talk to.

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Coreference Resolution

BT : Token Classification
TT : Token Classification

Coreference resolution is the task of identifying relationships between words or phrases in text that refer to the same entity.

Depth Estimation

EX : Model example: Cutting-edge depth estimation model.
Dataset example: NYU Depth V2 Dataset: Video dataset containing both RGB and depth sensor data.
Library example: transformers.
Demo example: An application that predicts the depth of an image and then reconstruct the 3D model as voxels.

Depth estimation is the task of predicting depth of the objects present in an image.

Dialogue Generation

BT : Text Generation
TT : Text Generation

Dialogue generation is the task of automatically generating dialogues.

Dialogue Modeling

BT : Text Generation
TT : Text Generation

Dialogue modeling is the task of organizing information in a conversation, where each event updates the current state of the dialogue based on the contents of the event.

Document Question Answering

EX : Dataset example: Largest document understanding dataset.
Library example: transformers.
Model example: A robust document question answering model.
Demo example: A robust document question answering application.

Document Question Answering (also known as Document Visual Question Answering) is the task of answering questions on document images. Document question answering models take a (document, question) pair as input and return an answer in natural language. Models usually rely on multi-modal features, combining text, position of words (bounding-boxes) and image.

Document Retrieval

BT : Text Retrieval
TT : Text Retrieval

Document retrieval is the task of matching some stated user query against a set of free-text records.

Entity Linking Classification

BT : Text Classification
TT : Text Classification

Entity linking classification is the task of predicting the correct unique identity for entities mentioned in text.

Entity Linking Retrieval

BT : Text Retrieval
TT : Text Retrieval

Entity linking retrieval is the task of retrieving and linking mentions of entities in unstructured text to corresponding entries in a structured knowledge base.

Explanation Generation

BT : Text Generation
TT : Text Generation

Explanation generation is the task of generating explanations for machine learning model predictions.

Extractive QA

BT : Question Answering
TT : Question Answering

Extractive question answering is the task of extracting the answer to a question from a given context.

Face Detection

BT : Object Detection
TT : Object Detection

Face detection is the task of detecting presence and location of faces.

Fact Checking

BT : Text Classification
TT : Text Classification

Fact checking is the task of verifying information in non-fictional text in order to determine its veracity and factual correctness.

Fact Checking Retrieval

BT : Text Retrieval
TT : Text Retrieval

Fact checking retrieval is the task of verifying information in non-fictional text to determine veracity and factual correctness by obtaining relevant information resources from a collection.

Feature Extraction

EX : Demo example: A leaderboard to rank text feature extraction models based on a benchmark.
Model example: A powerful feature extraction model for natural language processing tasks.
Dataset example: Wikipedia dataset containing cleaned articles of all languages. Can be used to train 'feature-extraction' models.
Library example: sentence-transformers.

Feature extraction is the task of extracting features learnt in a model.

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Fill Mask

EX : Dataset example: A common dataset that is used to train models for many languages.

Model example: State-of-the-art masked language model.

Library example: transformers.

NT : Masked Language Modeling

Masked language modeling is the task of masking some of the words in a sentence and predicting which words should replace those masks. These models are useful when we want to get a statistical understanding of the language in which the model is trained in.

Graph Machine Learning

NT : RDF-to-Text

Graph machine learning is a machine learning approach for graph data.

Grasping

BT : Robotics

TT : Robotics

Grasping is the task of automatically detecting and manipulating objects using robotic systems.

Hate Speech Detection

BT : Text Classification

TT : Text Classification

Hate speech detection is the task of detecting language used to express hatred, intolerance, or violence toward a particular group or individual based on characteristics such as race, ethnicity, nationality, religion, gender, or sexual orientation.

Image Captioning

BT : Image-to-Text

TT : Image-to-Text

Image captioning is the task of predicting a caption for a given image.

Image Classification

EX : Demo example: A leaderboard to evaluate different image classification models.

Model example: A strong image classification model.

Library example: keras.

Dataset example: Benchmark dataset used for image classification with images that belong to 100 classes.

NT : Multi Class Image Classification

Multi Label Image Classification

Image classification is the task of assigning a label or class to an entire image. Images are expected to have only one class for each image. Image classification models take an image as input and return a prediction about which class the image belongs to.

Image Colorization

BT : Image-to-Image

TT : Image-to-Image

Image colorization is the task of adding colors to grayscale images.

Image Feature Extraction

EX : Model example: A powerful image feature extraction model.

Dataset example: ImageNet-1K is a image classification dataset in which images are used to train image-feature-extraction models.

Library example: timm.

Demo example: A leaderboard to evaluate different image-feature-extraction models on classification performances

Image feature extraction is the task of extracting features learnt in a computer vision model.

Image Inpainting

BT : Image-to-Image

TT : Image-to-Image

Image inpainting is the task of filling erased or missing regions of an image.

Image Segmentation

EX : Dataset example: Scene segmentation dataset.

Demo example: A semantic segmentation application that can predict unseen instances out of the box.

Model example: Solid panoptic segmentation model trained on COCO.

Library example: transformers.

NT : Super Resolution

Panoptic Segmentation

Semantic Segmentation

Instance Segmentation

Image Segmentation divides an image into segments where each pixel in the image is mapped to an object. This task has multiple variants such as instance segmentation, panoptic segmentation and semantic segmentation.

Image Text-to-Image

EX : Demo example: An application for image-text-to-image generation.

Library example: diffusers.

Model example: A powerful model for image-text-to-image generation.

Image-text-to-image models take an image and a text prompt as input and generate a new image based on the reference image and text instructions. These models are useful for image editing, style transfer, image variations, and guided image generation tasks.

Image Text-to-Text

EX : Model example: Small and efficient yet powerful vision language model.

Demo example: Leaderboard to evaluate vision language models.

Library example: transformers.

Dataset example: Instructions composed of image and text.

Image-text-to-text models take in an image and text prompt and output text. These models are also called vision-language models, or VLMs. The difference from image-to-text models is that these models take an additional text input, not restricting the model to certain use cases like image captioning, and may also be trained to accept a conversation as input.

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Image Text-to-Video

EX : Library example: diffusers.

Model example: A powerful model for image-text-to-video generation.

Demo example: An application for image-text-to-video generation.

Image-text-to-video models take an reference image and a text instructions as and generate a video based on them. These models are useful for animating still images, creating dynamic content from static references, and generating videos with specific motion or transformation guidance.

Image-to-3D

EX : Model example: Fast image-to-3D mesh model by Tencent.

Library example: diffusers.

Demo example: Leaderboard to evaluate image-to-3D models.

Dataset example: A large dataset of over 10 million 3D objects.

Image-to-3D models take in image input and produce 3D output.

Image-to-Image

EX : Library example: diffusers.

Dataset example: Synthetic dataset, for image relighting

Demo example: Image editing application.

Model example: An image-to-image model to improve image resolution.

NT : Image Inpainting

Image Colorization

Image-to-image is the task of transforming an input image through a variety of possible manipulations and enhancements, such as super-resolution, image inpainting, colorization, and more.

Image-to-Text

EX : Library example: transformers.

Model example: Strong OCR model.

Dataset example: Dataset from 12M image-text of Reddit

Demo example: SVG generator app from images.

NT : Image Captioning

Image to text models output a text from a given image. Image captioning or optical character recognition can be considered as the most common applications of image to text.

Image-to-Video

EX : Library example: diffusers.

Dataset example: A benchmark dataset for reference image controlled video generation.

Demo example: An application to generate videos fast.

Model example: LTX-Video, a 13B parameter model for high quality video generation

Image-to-video models take a still image as input and generate a video. These models can be guided by text prompts to influence the content and style of the output video.

Instance Segmentation

BT : Object Detection

Image Segmentation

TT : Image Segmentation

Object Detection

Instance segmentation is the task of predicting exact pixel-wise boundaries of each individual object instance in an image.

Intent Classification

BT : Text Classification

TT : Text Classification

Intent classification is the task of deducing the specific purpose or objective behind a user's input in a conversational context.

Keypoint Detection

EX : Dataset example: A dataset of hand keypoints of over 500k examples.

Model example: A robust keypoint detection model.

Library example: transformers.

Demo example: An application that detects hand keypoints in real-time.

NT : Pose Estimation

Keypoint detection is the task of identifying meaningful distinctive points or features in an image.

Keyword Spotting

BT : Automatic Speech Recognition

TT : Automatic Speech Recognition

Keyword spotting is the task of identifying keywords in utterances.

Language Identification

BT : Text Classification

TT : Text Classification

Language identification is the task of determining language from a text sample.

Language Modeling

BT : Text Generation

TT : Text Generation

Language modeling is the task of predicting the next word in a sequence from a fixed size window of previous words.

Lemmatization

BT : Token Classification

TT : Token Classification

Lemmatization is the task of reducing inflected forms of words across text to a common root word or dictionary form, also called a lemma.

Masked Language Modeling

BT : Fill Mask

Text Generation

TT : Text Generation

Fill Mask

Masked language modeling is the task of predicting a masked word in a sentence.

Machine Learning Task Scheme : A-Z



Mask Generation

EX : Model example: Small yet powerful mask generation model.
Demo example: An application that combines a mask generation model with a zero-shot object detection model for text-guided image segmentation.
Dataset example: Widely used benchmark dataset for multiple Vision tasks.
Library example: transformers.

Mask generation is the task of generating masks that identify a specific object or region of interest in a given image. Masks are often used in segmentation tasks, where they provide a precise way to isolate the object of interest for further processing or analysis.

Multi Class Classification

BT : Text Classification
TT : Text Classification

Multi-class classification is the task of classifying instances into one of three or more classes.

Multi Class Image Classification

BT : Image Classification
TT : Image Classification

Multi-class image classification is the task of classifying images into one of three or more classes.

Multi Input Text Classification

BT : Text Classification
TT : Text Classification

Multi-input text classification is the task of predicting one or more labels for a given example based on multiple separate text inputs provided simultaneously.

Multi Label Classification

BT : Text Classification
TT : Text Classification

Multi-label classification is a variant of the classification problem where multiple labels may be assigned to each instance.

Multi Label Image Classification

BT : Image Classification
TT : Image Classification

Multi-label image classification is the task of assigning multiple relevant categories to an image.

Multiple Choice

NT : Tabular Multi Label Classification
Multiple Choice QA
Multiple Choice Coreference Resolution

A multiple choice task is similar to question answering, except several candidate answers are provided along with a context and the model is trained to select the correct answer.

Multiple Choice Coreference Resolution

BT : Multiple Choice
TT : Multiple Choice

Multiple choice coreference resolution is the task of finding expressions that refer to the same entity in text among several candidate answers provided.

Multiple Choice QA

BT : Question Answering
Multiple Choice
TT : Question Answering
Multiple Choice

Multiple choice question answering is the task of selecting the correct answer that correctly answers the question among several candidate answers provided along with a context.

Multivariate Time Series Forecasting

BT : Time Series Forecasting
TT : Time Series Forecasting

Multivariate time series forecasting is the task of predicting future values of multiple related variables by learning from their past behavior over time.

Named Entity Recognition

BT : Token Classification
TT : Token Classification

Named entity recognition is the task of extracting named entity mentions in unstructured text into pre-defined categories.

Natural Language Inference

BT : Text Classification
TT : Text Classification

Natural language inference is the task of classifying a pair of premise and hypothesis sentences into three classes: contradiction, neutral, and entailment.

News Articles Headline Generation

BT : Summarization
TT : Summarization

News articles headline generation is the task of generating a headline from the content of a news article.

News Articles Summarization

BT : Summarization
Question Answering
Text Classification
Text Generation
Token Classification
TT : Token Classification
Question Answering
Text Generation
Text Classification
Summarization

News article summarization is the task of creating a shorter version of a news article that captures all the important information.

Machine Learning Task Scheme : A-Z



Object Detection

EX : Dataset example: Widely used benchmark dataset for multiple vision tasks.
Model example: Solid object detection model pre-trained on the COCO 2017 dataset.
Demo example: Real-time object detection demo.
Library example: transformers.

NT : Panoptic Segmentation
Vehicle Detection
Instance Segmentation
Face Detection

Object Detection models allow users to identify objects of certain defined classes. Object detection models receive an image as input and output the images with bounding boxes and labels on detected objects.

Open Book QA

BT : Question Answering
Text Generation
TT : Question Answering
Text Generation

Open book question answering is the task of generating an answer directly based on context that correctly answers the question.

Open Domain Abstractive QA

BT : Question Answering
TT : Question Answering

Open domain abstractive question answering is the task of generating an answer from context that correctly answers the question without being restricted to a specific domain.

Open Domain QA

BT : Question Answering
TT : Question Answering

Open domain question answering is the task of generating or extracting an answer from context that correctly answers the question without being restricted to a specific domain.

Panoptic Segmentation

BT : Object Detection
Image Segmentation
TT : Image Segmentation
Object Detection

Panoptic segmentation is the task of segmenting an image into constituent parts, including both background regions and countable objects.

Parsing

BT : Token Classification
TT : Token Classification

Parsing is the task of analyzing and breaking down input data into component parts to extract meaningful information.

Part Of Speech

BT : Token Classification
TT : Token Classification

Part of speech is the task of identifying the grammatical category of each word in a sentence.

Pose Estimation

BT : Keypoint Detection
TT : Keypoint Detection

Pose estimation is the task of detecting and localizing keypoints on a human body, hand, or face in images and videos, outputting 2D or 3D coordinates.

Question Answering

EX : Library example: adapter-transformers.
Demo example: An application that can answer a long question from Wikipedia.
Dataset example: A famous question answering dataset based on English articles from Wikipedia.
Model example: A robust baseline model for most question answering domains.

NT : Open Domain Abstractive QA
Tabular Multi Label Classification
Closed Book QA
News Articles Summarization
Multiple Choice QA
Extractive QA
Open Domain QA
Open Book QA
Abstractive QA
Closed Domain QA

Question Answering models can retrieve the answer to a question from a given text, which is useful for searching for an answer in a document. Some question answering models can generate answers without context!

RDF-to-Text

BT : Graph Machine Learning
TT : Graph Machine Learning

RDF-to-text is the task of generating natural language from a triple representation in a knowledge graph.

Reinforcement Learning

EX : Dataset example: A curation of widely used datasets for Data Driven Deep Reinforcement Learning (D4RL)
Demo example: An application for a cute puppy agent learning to catch a stick.
Model example: A Reinforcement Learning model trained on expert data from the Gym Hopper environment
Library example: transformers.

Reinforcement learning is the computational approach of learning from action by interacting with an environment through trial and error and receiving rewards (negative or positive) as feedback

Robotics

NT : Grasping

Robotics is the task of improving the perception, decision-making, and interaction capabilities of robots.

Semantic Segmentation

BT : Image Segmentation
TT : Image Segmentation

Semantic segmentation is the task of assigning a class label to pixels.

Machine Learning Task Scheme : A-Z



Semantic Similarity Classification

BT : Text Classification

TT : Text Classification

Semantic similarity classification is the task of determining likeness in meaning between two linguistic elements, such as words or phrases, using predetermined discrete categories or labels.

Semantic Similarity Scoring

BT : Sentence Similarity

TT : Sentence Similarity

Semantic similarity scoring is the task of determining likeness in meaning between two linguistic elements, such as words or phrases, by assigning numeric scores.

Sentence Similarity

EX : Model example: This model works well for sentences and paragraphs and can be used for clustering/grouping and semantic searches.

Dataset example: Bing queries with relevant passages from various web sources.

Demo example: An application that leverages sentence similarity to answer questions from YouTube videos.

Library example: sentence-transformers.

NT : Semantic Similarity Scoring

Sentence Similarity is the task of determining how similar two texts are. Sentence similarity models convert input texts into vectors (embeddings) that capture semantic information and calculate how close (similar) they are between them. This task is particularly useful for information retrieval and clustering/grouping.

Sentiment Analysis

BT : Text Classification

TT : Text Classification

Sentiment analysis is the task of determining the emotional tone or sentiment behind a piece of text.

Sentiment Classification

BT : Text Classification

TT : Text Classification

Sentiment classification is the task of tagging data according to sentiment using predetermined discrete categories or labels.

Sentiment Scoring

BT : Text Classification

TT : Text Classification

Sentiment scoring is the task of tagging data according to sentiment by assigning numeric scores.

Slot Filling

BT : Token Classification

TT : Token Classification

Slot filling is the task of identifying meaningful slots in text.

Speaker Identification

BT : Automatic Speech Recognition

TT : Automatic Speech Recognition

Speaker identification is the task of determining the identity of an unknown speaker by comparing voice patterns with a database of known patterns, typically in a 1:N match.

Summarization

EX : Model example: A strong summarization model trained on English news articles. Excels at generating factual summaries.

Dataset example: News articles in five different languages along with their summaries. Widely used for benchmarking multilingual summarization models.

Demo example: An application that can summarize long paragraphs.

Library example: transformers.

NT : Text Simplification

News Articles Summarization

News Articles Headline Generation

Summarization is the task of producing a shorter version of a document while preserving its important information. Some models can extract text from the original input, while other models can generate entirely new text.

Super Resolution

BT : Other

Image Segmentation

TT : Other

Image Segmentation

Super resolution is the task of improving resolution of an imaging system beyond conventional limits.

Table Question Answering

EX : Demo example: An application that answers questions based on table CSV files.

Dataset example: The WikiTableQuestions dataset is a large-scale dataset for the task of question answering on semi-structured tables.

Library example: transformers.

Model example: A table question answering model that is capable of neural SQL execution, i.e., employ TAPEX to execute a SQL query on a given table.

NT : Tabular Multi Label Classification

Table Question Answering (Table QA) is the answering a question about an information on a given table.

Table-to-Text

Table-to-text is the task of converting structured tabular data into fluent and coherent natural language text.

Tabular Classification

EX : Demo example: An application that can predict defective products on a production line.

Dataset example: A comprehensive curation of datasets covering all benchmarks.

Model example: Breast cancer prediction model based on decision trees.

Library example: sklearn.

NT : Tabular Multi Label Classification

Tabular Multi Class Classification

Tabular classification is the task of classifying a target category (a group) based on set of attributes.

Tabular Multi Class Classification

BT : Tabular Classification

TT : Tabular Classification

Tabular multi-class classification is the task of classifying tabular data into one of three or more classes.

Machine Learning Task Scheme : A-Z



Tabular Multi Label Classification

BT : Question Answering
Text Classification
Text Generation
Tabular Classification
Multiple Choice
Zero Shot classification
Table Question Answering

TT : Question Answering
Text Generation
Text Classification
Tabular Classification
Multiple Choice
Zero Shot classification
Table Question Answering

Tabular multi-label classification is the task of assigning one or multiple labels to tabular data.

Tabular Regression

EX : Model example: Fish weight prediction based on length measurements and species.
Dataset example: A comprehensive curation of datasets covering all benchmarks.
Demo example: An application that can predict weight of a fish based on set of attributes.
Library example: sklearn.

NT : Tabular Single Column Regression

Tabular regression is the task of predicting a numerical value given a set of attributes.

Tabular Single Column Regression

BT : Tabular Regression
TT : Tabular Regression

Tabular single column regression is the task of predicting a numerical value given a single column of data.

Tabular-to-Text

Tabular-to-text is the task of converting structured tabular data into fluent and coherent natural language text.

Task Planning

BT : Text Generation
TT : Text Generation

Task planning is the task of automatically planning and scheduling realization of strategies or action sequences, typically for execution by intelligent agents, autonomous robots, and unmanned vehicles.

Text2text Generation

BT : Text Generation
TT : Text Generation

Text-to-text generation is the task of transforming input text into a different, task-specific output.

Text Classification

EX : Model example: A robust model trained for sentiment analysis.
Dataset example: A widely used dataset used to benchmark multiple variants of text classification.
Library example: adapter-transformers.
Demo example: An application that can classify financial sentiment.

NT : Multi Input Text Classification
Tabular Multi Label Classification
Multi Label Classification
Entity Linking Classification
Natural Language Inference
Text Scoring
Hate Speech Detection
News Articles Summarization
Sentiment Classification
Topic Classification
Semantic Similarity Classification
Fact Checking
Sentiment Scoring
Language Identification
Intent Classification
Sentiment Analysis
Multi Class Classification
Acceptability Classification

Text Classification is the task of assigning a label or class to a given text. Some use cases are sentiment analysis, natural language inference, and assessing grammatical correctness.

Text Generation

EX : Model example: A text-generation model trained to follow instructions.
Demo example: An application that writes and executes code from text instructions and supports many models.
Dataset example: Multilingual dataset used to evaluate text generation models.
Library example: transformers.

NT : Explanation Generation
Dialogue Generation
Tabular Multi Label Classification
Language Modeling
Conversational
Dialogue Modeling
Closed Book QA
News Articles Summarization
Text2text Generation
Open Book QA
Task Planning
Masked Language Modeling

Generating text is the task of generating new text given another text. These models can, for example, fill in incomplete text or paraphrase.

Text Ranking

EX : Model example: An extremely efficient text ranking model trained on a web search dataset.
Dataset example: Bing queries with relevant passages from various web sources.
Library example: sentence-transformers.

Text Ranking is the task of ranking a set of texts based on their relevance to a query. Text ranking models are trained on large datasets of queries and relevant documents to learn how to rank documents based on their relevance to the query. This task is particularly useful for search engines and information retrieval systems.

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Text Retrieval

NT : Entity Linking Retrieval
Document Retrieval
Utterance Retrieval
Fact Checking Retrieval

Text retrieval is the task of obtaining information resources relevant to an information need from text.

Text Scoring

BT : Text Classification
TT : Text Classification

Text scoring is the task of assigning a numerical or categorical score to a piece of text to evaluate its quality, relevance, or readability.

Text Simplification

BT : Summarization
TT : Summarization

Text simplification is the task of modifying, organizing, or categorizing existing text to make it easier to understand while retaining its original meaning.

Text-to-3D

EX : Library example: diffusers.
Demo example: Text-to-3D demo with mesh outputs.
Model example: Text-to-3D mesh model by OpenAI
Dataset example: A large dataset of over 10 million 3D objects.

Text-to-3D models take in text input and produce 3D output.

Text-to-Audio

Text-to-audio, also known as text-to-speech (TTS), is the task of generating natural sounding speech given text input.

Text-to-Image

EX : Library example: diffusers.
Dataset example: RedCaps is a large-scale dataset of 12M image-text pairs collected from Reddit.
Model example: One of the most powerful image generation models that can generate realistic outputs.
Demo example: A powerful text-to-image application.

Text-to-image is the task of generating images from input text. These pipelines can also be used to modify and edit images based on text prompts.

Text-to-Speech

EX : Model example: Small yet powerful TTS model.
Dataset example: 10K hours of multi-speaker English dataset.
Demo example: An application for generate high quality speech in different languages.
Library example: espnet.

Text-to-Speech (TTS) is the task of generating natural sounding speech given text input. TTS models can be extended to have a single model that generates speech for multiple speakers and multiple languages.

Text-to-Video

EX : Library example: diffusers.
Demo example: An application that generates video from text.
Dataset example: Microsoft Research Video to Text is a large-scale dataset for open domain video captioning
Model example: A strong model for consistent video generation.

Text-to-video models can be used in any application that requires generating consistent sequence of images from text.

Time Series Forecasting

NT : Univariate Time Series Forecasting
Multivariate Time Series Forecasting

Time series forecasting is the task of predicting future values based on previously observed values.

Token Classification

EX : Dataset example: A widely used dataset useful to benchmark named entity recognition models.
Demo example: An application that can recognizes entities, extracts noun chunks and recognizes various linguistic features of each token.
Library example: adapter-transformers.
Model example: A robust performance model to identify people, locations, organizations and names of miscellaneous entities.
NT : Named Entity Recognition
Part Of Speech
News Articles Summarization
Coreference Resolution
Parsing
Lemmatization
Slot Filling
Word Sense Disambiguation

Token classification is a natural language understanding task in which a label is assigned to some tokens in a text. Some popular token classification subtasks are Named Entity Recognition (NER) and Part-of-Speech (PoS) tagging. NER models could be trained to identify specific entities in a text, such as dates, individuals and places; and PoS tagging would identify, for example, which words in a text are verbs, nouns, and punctuation marks.

Topic Classification

BT : Text Classification
TT : Text Classification

Topic classification is the task of assigning predicted labels representing a piece of text's overall primary set of topics to this piece of text.

Translation

EX : Model example: Very powerful model that can translate many languages between each other, especially low-resource languages.
Library example: transformers.
Dataset example: A dataset of copyright-free books translated into 16 different languages.
Demo example: An application that can translate between 100 languages.

Translation is the task of converting text from one language to another.

Machine Learning Task Scheme : A-Z



Unconditional Image Generation

EX : Library example: diffusers.

Demo example: An application that can generate realistic faces.

Dataset example: The CIFAR-100 dataset consists of 60000 32x32 colour images in 100 classes, with 600 images per class.

Model example: High-quality image generation model trained on the CIFAR-10 dataset. It synthesizes images of the ten classes presented in the dataset using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics.

Unconditional image generation is the task of generating images with no condition in any context (like a prompt text or another image). Once trained, the model will create images that resemble its training data distribution.

Univariate Time Series Forecasting

BT : Time Series Forecasting

TT : Time Series Forecasting

Univariate time series forecasting is the task of making predictions for a single variable using available historical data points for that variable.

Utterance Retrieval

BT : Text Retrieval

TT : Text Retrieval

Utterance retrieval is the task of obtaining information resources relevant to an information need from an utterance.

Vehicle Detection

BT : Object Detection

TT : Object Detection

Vehicle detection is the task of detecting vehicles.

Video Classification

EX : Demo example: An application that classifies video at different timestamps.

Dataset example: Benchmark dataset used for video classification with videos that belong to 400 classes.

Library example: transformers.

Model example: Strong Video Classification model trained on the Kinetics 400 dataset.

Video classification is the task of assigning a label or class to an entire video. Videos are expected to have only one class for each video. Video classification models take a video as input and return a prediction about which class the video belongs to.

Video Text-to-Text

EX : Demo example: An application to chat with a video-text-to-text model.

Model example: A robust video-text-to-text model.

Library example: transformers.

Dataset example: Multiple-choice questions and answers about videos.

Video-text-to-text models take in a video and a text prompt and output text. These models are also called video-language models.

Video-to-Video

EX : Dataset example: Dataset with detailed annotations for training and benchmarking video instance editing.

Library example: diffusers.

Model example: Model for editing outfits, character, and scenery in videos.

Demo example: Interactive demo space for Lucy-Edit-Dev video editing.

Video-to-video models take one or more videos as input and generate new videos as output. They can enhance quality, interpolate frames, modify styles, or create new motion dynamics, enabling creative applications, video production, and research.

Visual Document Retrieval

EX : Model example: Very accurate visual document retrieval model for multilingual queries and documents.

Dataset example: A large dataset used to train visual document retrieval models.

Library example: transformers.

Demo example: A leaderboard of visual document retrieval models.

Visual document retrieval is the task of searching for relevant image-based documents, such as PDFs. These models take a text query and multiple documents as input and return the top-most relevant documents and relevancy scores as output.

Visual Question Answering

EX : Model example: A visual question answering model trained to convert charts and plots to text.

Library example: transformers.

Demo example: An application that compares visual question answering models across different tasks.

Dataset example: A widely used dataset containing questions (with answers) about images.

Visual Question Answering is the task of answering open-ended questions based on an image. They output natural language responses to natural language questions.

Voice Activity Detection

Voice activity detection is the task of detecting the presence or absence of human speech.

Word Sense Disambiguation

BT : Token Classification

TT : Token Classification

Word sense disambiguation is the task of identifying which sense of a word with multiple meanings is used in a sentence.

Zero Shot classification

EX : Dataset example: A widely used dataset used to benchmark multiple variants of text classification.

Model example: Powerful zero-shot text classification model.

Library example: transformers.

NT : Tabular Multi Label Classification

Zero-shot text classification is a task in natural language processing where a model is trained on a set of labeled examples but is then able to classify new examples from previously unseen classes.

Machine Learning Task Scheme : A-Z



Zero Shot image classification

EX : Model example: Multilingual image classification model for 80 languages.

Library example: transformers.

Demo example: An application that leverages zero-shot image classification to find best captions to generate an image.

Zero-shot image classification is the task of classifying previously unseen classes during training of a model.

Zero Shot object detection

EX : Model example: Solid zero-shot object detection model.

Demo example: A demo to compare different zero-shot object detection models per output and latency.

Library example: transformers.

Zero-shot object detection is a computer vision task to detect objects and their classes in images, without any prior training or knowledge of the classes. Zero-shot object detection models receive an image as input, as well as a list of candidate classes, and output the bounding boxes and labels where the objects have been detected.